

AOUANOUK Slimane

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Available for a 6-month internship starting February 2027 in robotics or autonomous systems.

Graduate Engineering student at Arts et Métiers and incoming Research Master's (M2) student in Advanced Systems and Robotics (SAR) at Sorbonne Université. Research experience in **real-time robotics, trajectory optimization, physical validation, and simulation-to-real transfer.** Currently studying **machine learning** (deep learning, reinforcement learning, PyTorch), **state estimation, and robot perception.** Background in **physics-based modeling, Python, experimental analysis, and system-level engineering.** Driven by **learning fast, building real systems, and turning ideas into structured experiments with defensible conclusions.**

EDUCATION — Combined GPA : 3.92 / 4.00

Sorbonne Université — Paris, France

Sept 2026 – Aug 2027

Research Master's (M2) in Advanced Systems & Robotics (SAR)

Selected research-oriented coursework: Mobile Robotics (ROS 2, SLAM), Robot Perception (camera-based vision and multi-sensor fundamentals), Machine Learning (deep learning, CNNs/RNNs, autoencoders, MDPs, reinforcement learning, DQN/DDPG, PyTorch), State Estimation & System Identification (Bayesian estimation, EKF, neural extended Kalman filter (NEKF), UKF, particle filters, and data-driven identification), Nonlinear Control, and Robot Manipulators.

Arts et Métiers Institute of Technology (ENSAM) — Paris, France

Sept 2024 – Jul 2027

Combined BS × MS Engineering Degree — Member of **ParisTech**; rigorous multidisciplinary training supported by world-class research laboratories.

- **Ranked top 12% of cohort (142 / 1223 students)**
- **Relevant Coursework:** Solid Mechanics, Robotics and Mechatronics, Numerical Optimization, Computer-Aided Design (CAD), Experimental Mechanics and Validation Testing, Fluid Mechanics

WORK EXPERIENCE

Trajectory Optimization and Collision Avoidance for Autonomous Systems 

Jan–Aug 2026

Robotics Research Project & Internship — PIMM Lab / CNRS@CREATE; Paris (Jan–May) → Singapore (Jun–Aug) — Supervisor: Prof. Francisco Chinesta

- Co-developed **end-to-end** and physically validated a **mobile-robot autonomy stack** — from trajectory planning and perception through real-time control — achieving **40/40 goal arrivals with zero collisions** across two frozen 20-run experimental campaigns
- Built a **closed-loop perception-to-action pipeline** integrating computer vision, real-time trajectory replanning, robot control, and **structured experiment logging**, sustaining a median **19.84–19.86 Hz** control loop
- **Diagnosed and resolved a perception failure mode** — reducing mean localization error by **84%** (36.3 → 5.8 mm) — through systematic lens-distortion correction and homography-based metric calibration
- Designed **structured experiments** comparing variational planners and controllers across **367,374 simulation runs** and a **100,800-case campaign**; quantified **simulation-to-real gap** across 39 paired runs and contributed to a **preprint draft in preparation**

SELECTED PROJECTS

AI-Driven Smart Workshop Project  — Arts et Métiers

2025

- Built an **end-to-end local data & AI pipeline** to query **technical workshop data** in **natural language**; **structured heterogeneous CSV data** and migrated it to a **relational SQLite database**; **validated and explored data** using **Python** and **pandas**; **designed SQL views** to simplify **complex joins** and improve **robustness**; **implemented a secure Text-to-SQL system** powered by a **locally deployed LLM (Ollama)** with **strict query constraints**; **developed a Streamlit interface** enabling **real-time usage** and **human-readable responses**

Plastic Collapse Analysis  — Arts et Métiers

2026

- Predicted a portal-frame collapse load within **2.08%** of physical measurement (**423 N** analytical vs **432 N** measured) by combining **limit analysis**, **Python-scripted Abaqus FEA models**, and **MTS 100 kN testing**; evaluated 1D and 2D collapse mechanisms across four load angles, documenting both model agreement and unresolved discrepancies

Hydraulic Turbine-Generator  — Arts et Métiers

2025–26

- Co-developed a turbine concept through hydraulic design, **CFD (STAR-CCM+)**, electromagnetic sizing, and **additive manufacturing**; simulated a 110–205 kg/s sweep on a **96,045-cell model** identifying a **75.6% peak efficiency**; manufactured a **150 mm PLA prototype** and measured **Cp≈0.28** in wind-tunnel testing

SKILLS AND INTERESTS

Programming & ML: Python (data analysis, numerical computation, scientific optimization), PyTorch (current coursework), MATLAB, SQL/SQLite, LaTeX

Robotics & Vision: Multi-robot hardware integration, ArUco localization, camera calibration, real-time trajectory testing, ROS 2 (current coursework)

Modeling & Simulation: CFD (STAR-CCM+), FEA (Abaqus), CAD (Fusion 360, 3DEXPERIENCE), Simulink, digital twins

Languages: French (native), English (TOEFL iBT: 5/6), Italian (basic), Spanish (basic)

Interests: Robot learning, physical AI, applied reinforcement learning, basketball, strength training